

Mike Hosian Maeda

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EDUCATION

Alfred University

Alfred, NY

Bachelor of Arts in Data Analytics & Computer Science - Double Major | GPA: 3.9/4.0

Expected May 2028

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Statistics, Calculus I
- **Honors & Awards:** Presidential Scholarship Recipient, Honors Program, Dean's List, AHA! Startup Competition Winner (\$5,000), Omicron Delta Kappa Leadership Honor Society

TECHNICAL SKILLS

Languages: Java, Python, SQL, JavaScript, TypeScript, C, C++, C#, Scala, HTML/CSS

Technologies: Flask, FastAPI, React, SQLite, MySQL, PostgreSQL, MongoDB, REST APIs, Git, Docker, Azure, Linux

EXPERIENCE

Undergraduate Research Intern

May 2026 – Present

Alfred University Inamori School of Engineering, (AI and Machine Learning)

Alfred, NY

- Developing a **computer vision-based e-waste classification system** using machine learning models to identify recyclable materials from electronic components, aiming to improve material sorting efficiency by **40%** compared to manual classification methods.
- Building automated **image processing and data analysis pipelines** to extract material features from e-waste samples, reducing manual analysis time by **50%** and improving consistency of recovery decisions.
- Collaborating with materials engineering researchers to integrate AI models into recycling workflows, supporting scalable recovery of critical materials including copper from complex electronic waste streams.

Robotics Software Engineer

Aug 2023 – May 2025

FIRST Robotics Challenge, (Java, Control Systems, Sensor Integration)

Ramat HaSharon, Israel

- Designed and implemented **Java-based autonomous robot control systems** using encoder feedback, IMU sensors, and control algorithms, improving navigation accuracy by **35%** through iterative testing and calibration.
- Developed real-time sensor feedback loops integrating gyroscope and encoder data to optimize drivetrain responsiveness, reducing autonomous movement errors by **30%** during competitive robot operations.

PROJECTS

School Management System(SMS) | Java, SQL

Oct 2025 – Nov 2025

- Engineered a full-stack **Java-based desktop information system to manage 1,000+** academic records, attendance tracking, improving administrative processing efficiency by **40%** through workflow automation and structured data management.
- Designed and implemented a **normalized relational SQLite database** with automated **SQL CRUD operations**, enforcing data integrity, reducing redundancy, and enabling scalable data storage as record volume increased.

Expense Tracker | Python, Flask, SQLite, OAuth, REST APIs, Vercel

April 2026

- Developed and deployed a **full-stack Flask financial management platform** with **8+ core features** including Google OAuth authentication, secure user sessions, expense tracking, budget management, and automated account notifications.
- Designed a **relational SQLite database system** with multiple interconnected tables supporting users, expenses, categories, and budgets, implementing **20+ CRUD operations** for scalable financial data management.
- Integrated **3 production-ready services** including SMTP email automation, CSV data export functionality, and Vercel cloud deployment with environment-based configuration and REST API architecture.

Statistical Analysis of NYC Coffee Shop Sales | Python, Excel, SQL, Statistics, Data Visualization

Nov 2025 – Dec 2025

- Analyzed **17,000+ real-world retail transactions** from Kaggle using regression modeling and statistical feature analysis, identifying a positive relationship between unit price and transaction value (**$R^2 = 0.43$**) supporting premium pricing strategy
- Performed data cleaning, transformation, and exploratory feature analysis across multi-variable retail datasets, improving dataset consistency and supporting reproducible statistical modeling workflows.

LEADERSHIP AND PROFESSIONAL DEVELOPMENT

Resident Assistant

May 2026 – Present

Alfred University Residence Life

Alfred, NY

- Support a residential community through peer mentorship, conflict resolution, event planning, and initiatives promoting student engagement and academic success.

Alumni Relations and Outreach Chair

Jan 2026 – Present

ColorStack Alfred University Chapter

Alfred, NY

- Lead outreach and professional development initiatives connecting underrepresented computing students with mentorship, networking, and technology career opportunities.